Power Quality Analyzer ITA-9000

Variant: [4G LTE/WIFI]

07/25/2025

V1.0

RELEASED

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DESIGN CONSIDERATIONS

DESIGN NOTE:
Example text for informational design notes.

DESIGN NOTE:
Example text for cautionary design notes.

DESIGN NOTE:
Example text for debug notes.

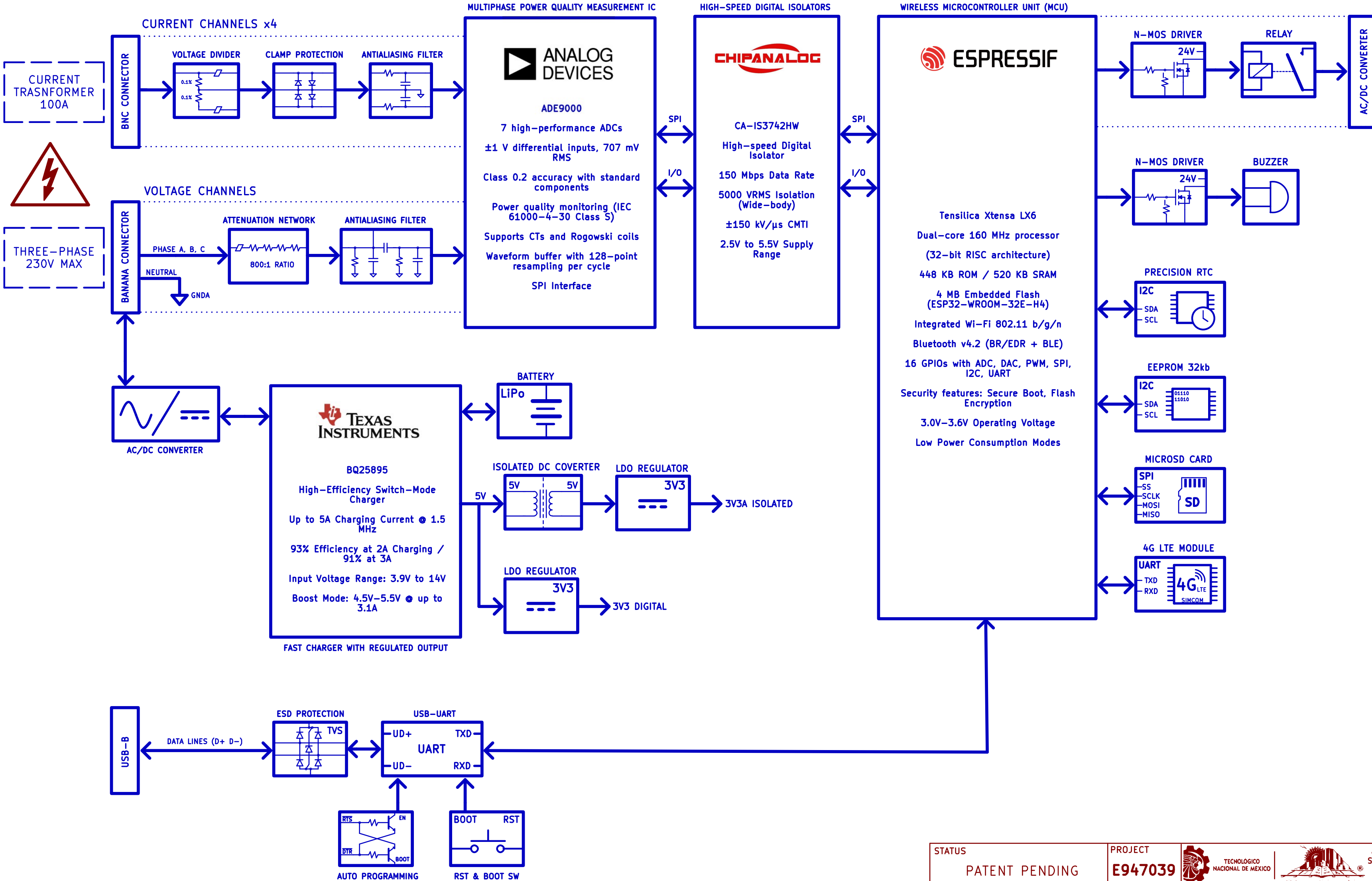
DESIGN NOTE:
Example text for critical design notes.

LAYOUT NOTE:
Example text for critical layout guidelines.

This project was developed as part of the master's thesis by Eng. Brayan Daniel Vazquez Gasca (CVU: 2064662), at Instituto Tecnologico de Apizaco. It was supported by a national postgraduate scholarship provided by Secretariat of Science, Humanities, Technology and Innovation (SECIHTI) of Mexico. Project thesis supervision by Dr. Roberto Morales Caporal

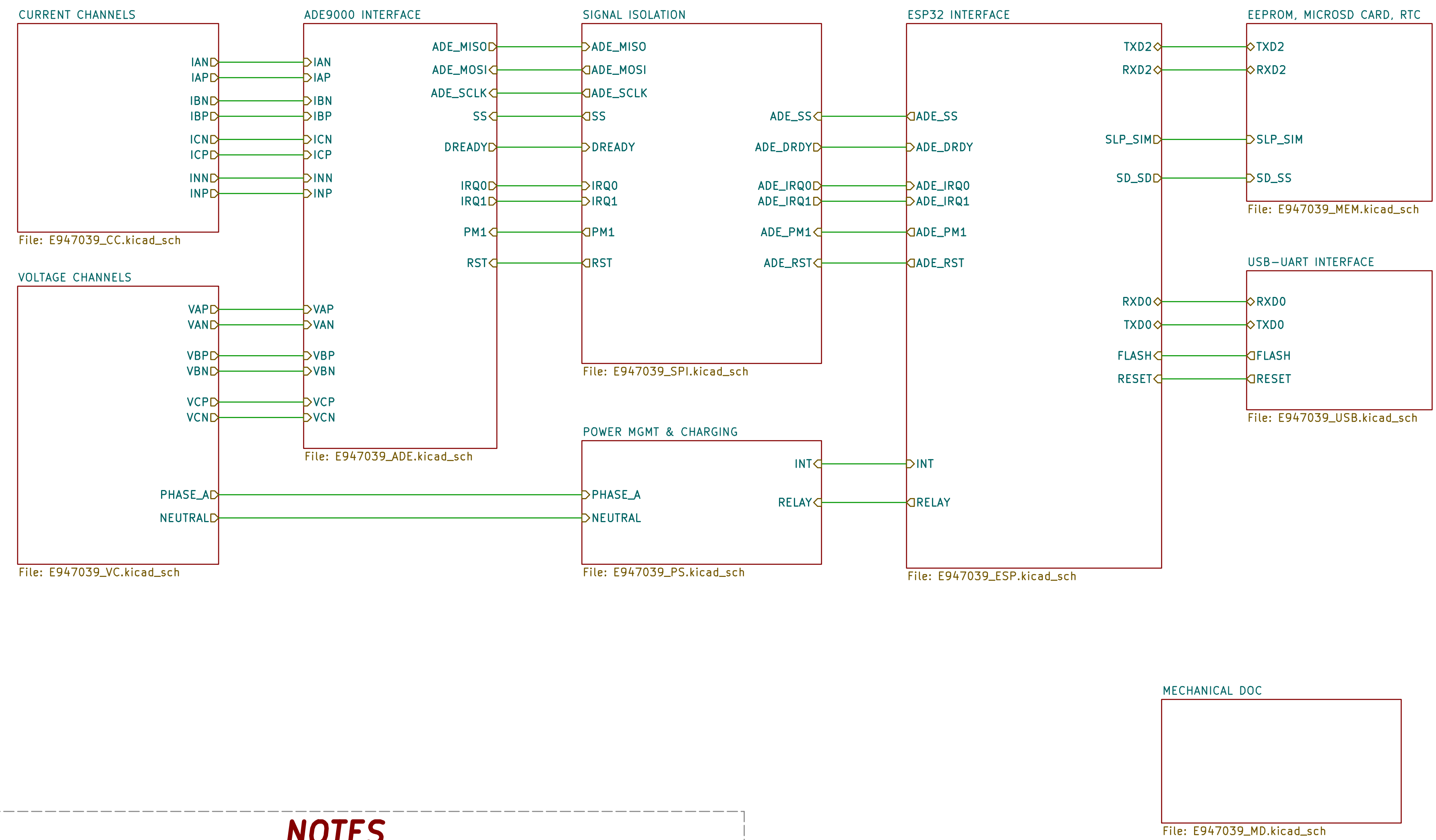
STATUS: PATENT PENDING	PROJECT: E947039	TECNOLÓGICO NACIONAL DE MÉXICO	Av. Instituto Tecnológico No. 418 San Andrés Ahuashuatepec Tzomp. Tlaxcala, México. C.P 90491 Tel: 241-417-2010 Ext.145
CONFIDENTIAL	PROJECT REVISION: A1	DESIGN ITEM: E947039	
APPROVALS	DATE	TITLE: Power Quality Analyzer ITA-9000	
ENG: Vazquez Gasca B. D.	07/21/25	SIZE: USLetter	DWG NO. 25GSK947039-A1 / REV: 1.0
RV1: Morales Caporal R.	07/21/25		
RV2:		SCALE: 1:1 / FILE NAME: E947039.kicad_sch	SHEET: A1/13

BLOCK DIAGRAM



STATUS: PATENT PENDING	PROJECT: E947039	TECNOLÓGICO NACIONAL DE MÉXICO	Av. Instituto Tecnológico No. 418, San Andrés Ahuashuatepec Tzomp. Tlaxcala, México. C.P 90491 Tel: 241-417-2010 Ext.145
CONFIDENTIAL	PROJECT REVISION: A1	DESIGN ITEM: BLOCK DIAGRAM	
APPROVALS / DATE	TITLE: *Power Quality Analyzer ITA-9000*		
ENG: Vazquez Gasca B. D. — 07/21/25	SIZE: USLetter	DWG NO.: 25GSK947039-A2	REV: 1.0
RV1: Morales Caporal R. — 07/21/25	SCALE: 1:1	FILE NAME: E947039_BD.kicad_sch	SHEET: A2/13
RV2:			

HIERARCHICAL SCHEMATIC



NOTES

DRAFT – Very early stage of schematic, ignore details.

PRELIMINARY – Close to final schematic.

CHECKED – There should not be any mistakes. Tell the engineer if you find one.

RELEASED – A board with this schematic has been sent to production.

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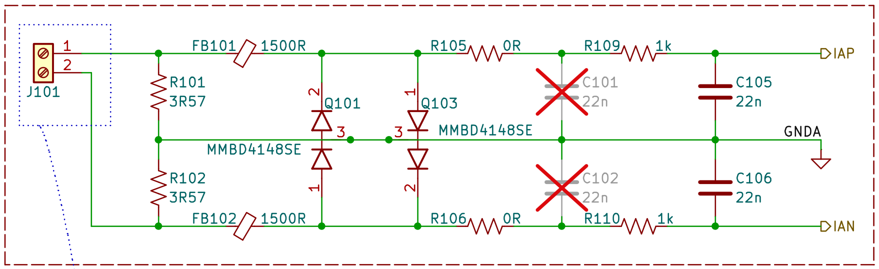
CONFIDENTIAL	PROJECT REVISION: A1	DESIGN ITEM: HIERARCHICAL SCHEMATIC	
APPROVALS	DATE	TITLE: Power Quality Analyzer ITA-9000	
ENG: Vazquez Gasca B. D.	07/21/25	SIZE: USLetter	DWG NO.: 25GSK947039-A3 — REV: 1.0
RV1: Morales Caporal R.	07/21/25	SCALE: 1:1	FILE NAME: E947039_HS.kicad_sch — SHEET: A3/13
RV2:			

CURRENT CHANNELS

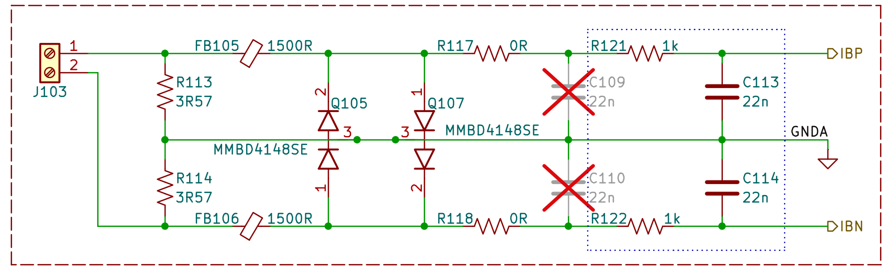
DESIGN NOTE:
This equipment is connected to hazardous line voltages. Exercise proper caution when connecting the sensors and voltage leads. Ensure that the system is enclosed in a protective casing.

DESIGN NOTE:
All Antialising capacitors are C0G/NP0 grade.

Phase A



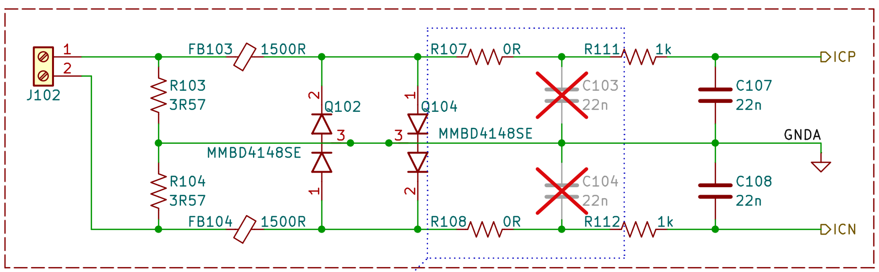
Phase B



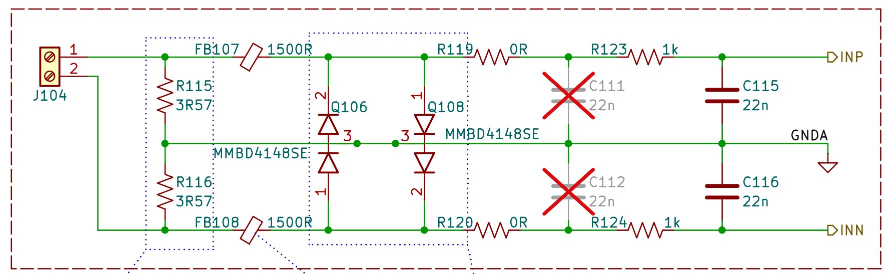
DESIGN NOTE:
The ITA-9000 is designed to operate directly with 100A/50mA output current transformers (CTs). Connect the CT cables to the terminal blocks J101, J102, J103, and J104.

DESIGN NOTE:
The antialiasing filter corner is chosen around 7 kHz to provide sufficient attenuation of out of band signals near the modulator clock frequency. The same RC filter corner is used on voltage channels as well, to avoid phase errors between current and voltage signals

Phase C



Neutral



DESIGN NOTE:
Not populated capacitors C1, C2, C3, C4, C9, C10, C11 and C12 allow switching to Rogowski coils instead of CT's. For more information check the user guide.

DESIGN NOTE:
To allow headroom, the input signal into the current channel analog-to-digital converter (ADC) at maximum current is set at half of full scale. Because the full-scale differential input is ±0.707 V rms, the total burden resistor, RB be calculated as the manual shows.

DESIGN NOTE:
Protection diodes to limit the voltage on the INP/INN and prevent damage from overvoltages of the CT's. The INP pin to GNDA pin and INN pin to GNDA pin potentials do not exceed ±0.7 V peak.

DESIGN NOTE:
Ferrite beads filters any high frequency noise present on the wires.

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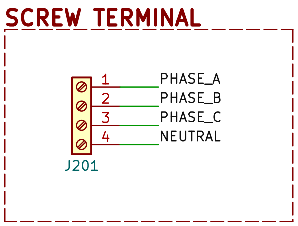
CONFIDENTIAL	PROJECT REVISION: A1	DESIGN ITEM: CURRENT CHANNELS
APPROVALS / DATE	TITLE: *Power Quality Analyzer ITA-9000*	
ENG: Vazquez Gasca B. D. — 07/21/25	SIZE: USLetter	DWG NO. 25GSK947039-01 — REV: 1.0
RV1: Morales Caporal R. — 07/21/25		
RV2:	SCALE: 1:1 — FILE NAME: E947039_CC.kicad_sch	SHEET: 1/13

VOLTAGE CHANNELS

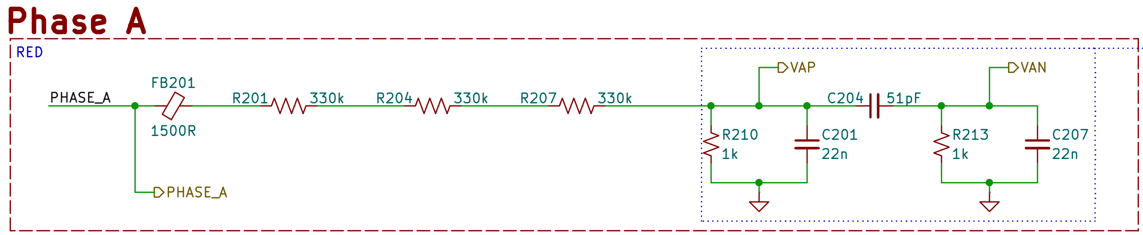
DESIGN NOTE:
This equipment is connected to hazardous line voltages. Exercise proper caution when connecting the sensors and voltage leads. Ensure that the system is enclosed in a protective casing.

DESIGN NOTE:
All Antialising capacitors are C0G/NP0 grade.

SCREW TERMINAL

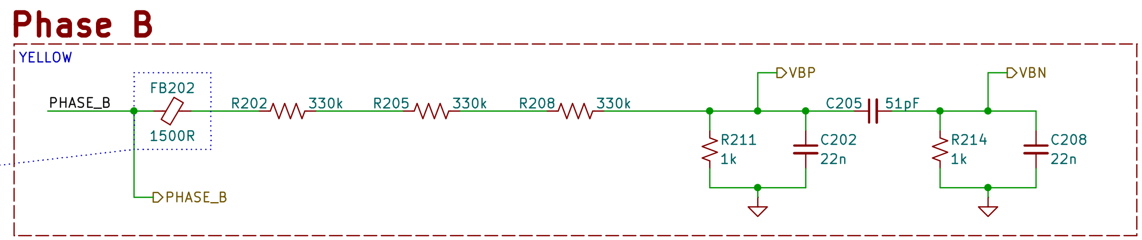


Phase A



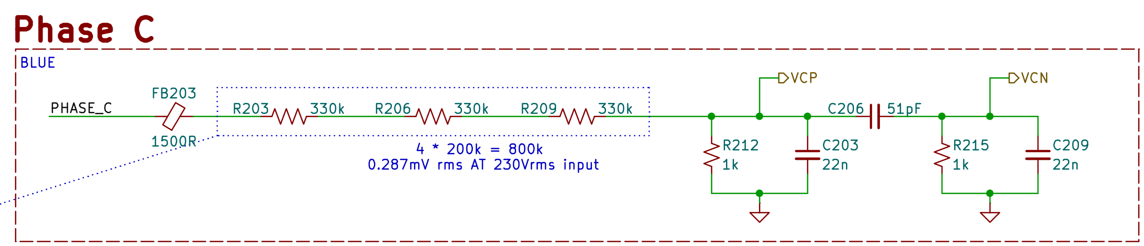
DESIGN NOTE:
The antialiasing filter corner is chosen around 7 kHz to provide sufficient attenuation of out of band signals near the modulator clock frequency. The same RC filter corner is used on current channels as well, to avoid phase errors between current and voltage signals.

Phase B



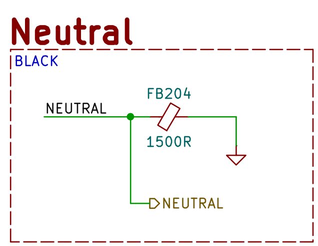
DESIGN NOTE:
Ferrite beads that filter any high frequency noise present on the wires.

Phase C



DESIGN NOTE:
There are three 330 kΩ resistors connected in series, forming an attenuation network with a 1 kΩ resistor. This setup provides an attenuation ratio of 991:1.

Neutral



DESIGN NOTE:
The neutral voltage input connects to AGND, establishing the ground reference for three-phase voltage measurements.

LAYOUT NOTE:
Do not place copper planes in this block. Maintain a minimum separation of [X mm] to ensure insulation.

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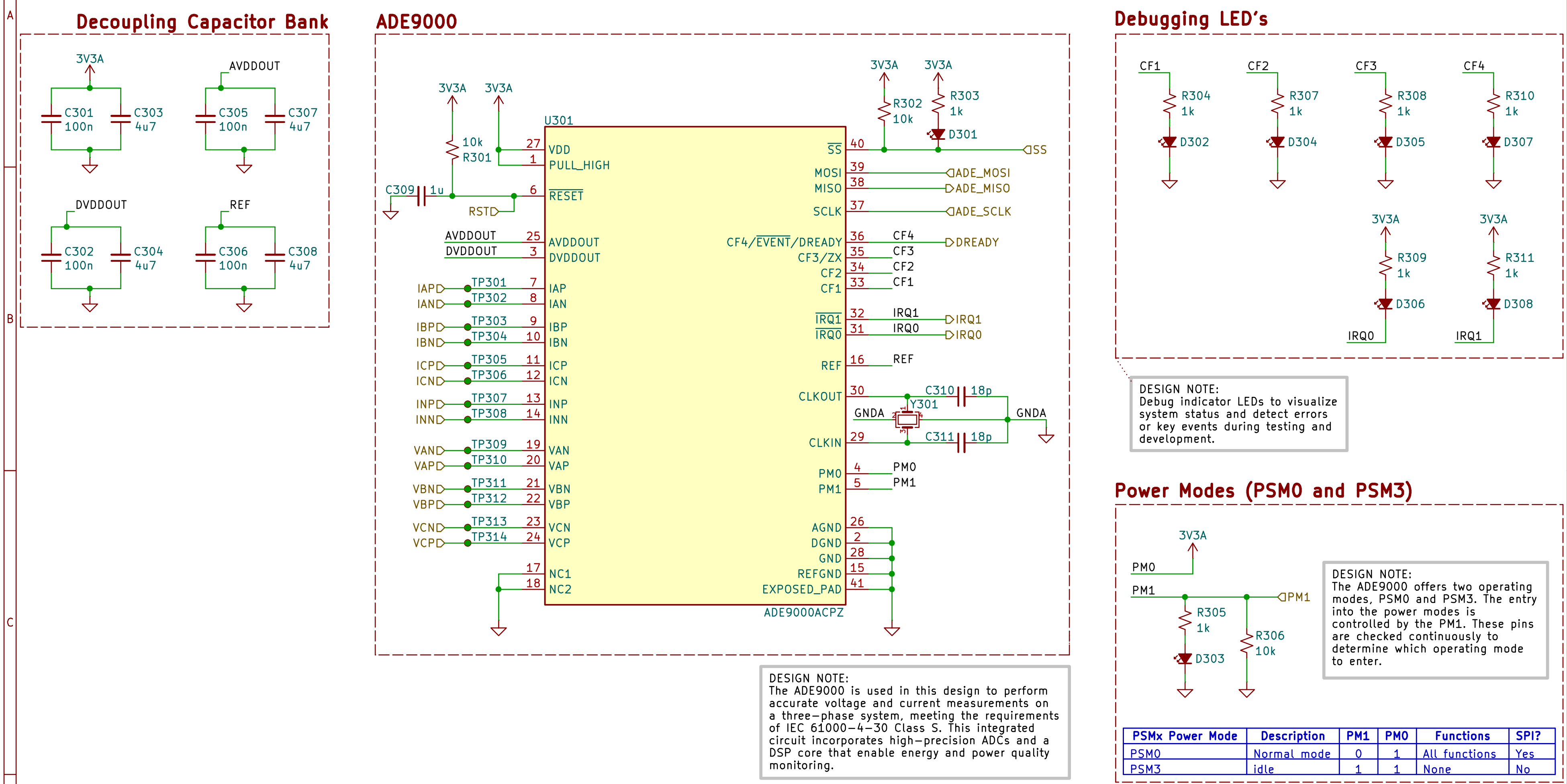
CONFIDENTIAL	PROJECT REVISION: A1	DESIGN ITEM: VOLTAGE CHANNELS	
APPROVALS	DATE	TITLE: Power Quality Analyzer ITA-9000	
ENG: Vazquez Gasca B. D.	07/21/25	SIZE: USLetter	DWG NO. 25GSK947039-02 — REV: 1.0
RV1: Morales Caporal R.	07/21/25	SCALE: 1:1	FILE NAME: E947039_VC.kicad_sch — SHEET: 2/13
RV2:			

ADE9000 INTERFACE



DESIGN NOTE:
Debug indicator LEDs to visualize system status and detect errors or key events during testing and development.

DESIGN NOTE:
The ADE9000 offers two operating modes, PSM0 and PSM3. The entry into the power modes is controlled by the PM1. These pins are checked continuously to determine which operating mode to enter.

PSMx Power Mode	Description	PM1	PM0	Functions	SPI?
PSM0	Normal mode	0	1	All functions	Yes
PSM3	idle	1	1	None	No

DESIGN NOTE:
The ADE9000 is used in this design to perform accurate voltage and current measurements on a three-phase system, meeting the requirements of IEC 61000-4-30 Class S. This integrated circuit incorporates high-precision ADCs and a DSP core that enable energy and power quality monitoring.

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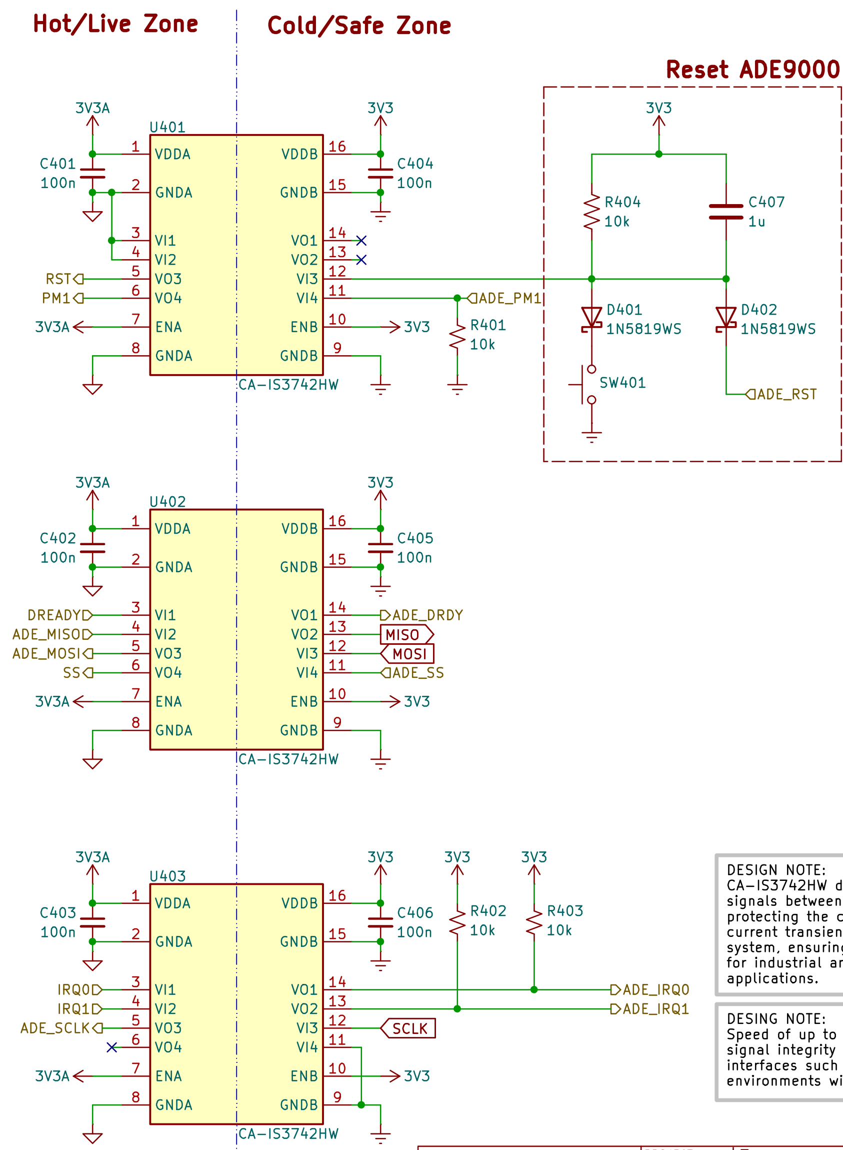
CONFIDENTIAL	PROJECT REVISION: A1	DESIGN ITEM: ADE9000 INTERFACE	
APPROVALS / DATE	TITLE: Power Quality Analyzer ITA-9000		
ENG: Vazquez Gasca B. D. / 07/21/25	SIZE: USLetter	DWG NO.: 25GSK947039-03	REV: 1.0
RV1: Morales Caporal R. / 07/21/25			
RV2:	SCALE: 1:1	FILE NAME: E947039_ADE.kicad_sch	SHEET: 3/13

SIGNAL ISOLATION



DESIGN NOTE:
CA-IS3742HW devices are used to isolate digital signals between independent ground domains, protecting the control logic from high voltage and current transients present in the measurement system, ensuring isolation up to 5kVRMS, suitable for industrial and three-phase power monitoring applications.

DESING NOTE:
Speed of up to 150Mbps by the isolators allow signal integrity to be maintained on fast interfaces such as SPI or UART, even in environments with high electrical noise.

LAYOUT NOTE:
Place 2mm slots/cutouts in half of the signal isolation ICs

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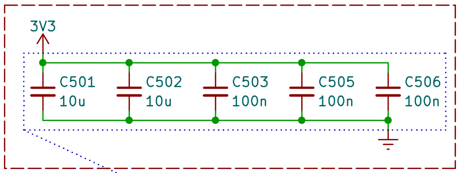
CONFIDENTIAL	PROJECT REVISION: A1	DESIGN ITEM: SIGNAL ISOLATION	
APPROVALS / DATE	TITLE: Power Quality Analyzer ITA-9000		
ENG: Vazquez Gasca B. D. / 07/21/25	SIZE: USLetter	DWG NO. 25GSK947039-04	REV: 1.0
RV1: Morales Caporal R. / 07/21/25			
RV2:	SCALE: 1:1	FILE NAME: E947039_SPI.kicad_sch	SHEET: 4/13

GPIO0, GPIO2, GPIO12 (MTDI) and GPIO15 (MTDO)

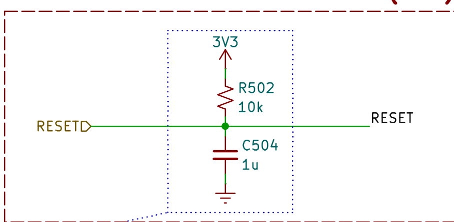
ESP32 INTERFACE

Decoupling Capacitor Bank



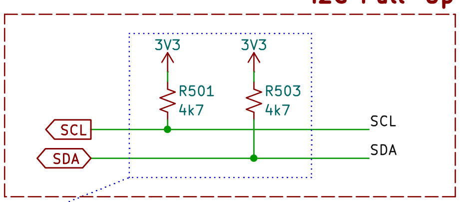
LAYOUT NOTE:
Decoupling capacitors less than 10mm far from power pins

Power-On Reset (POR)



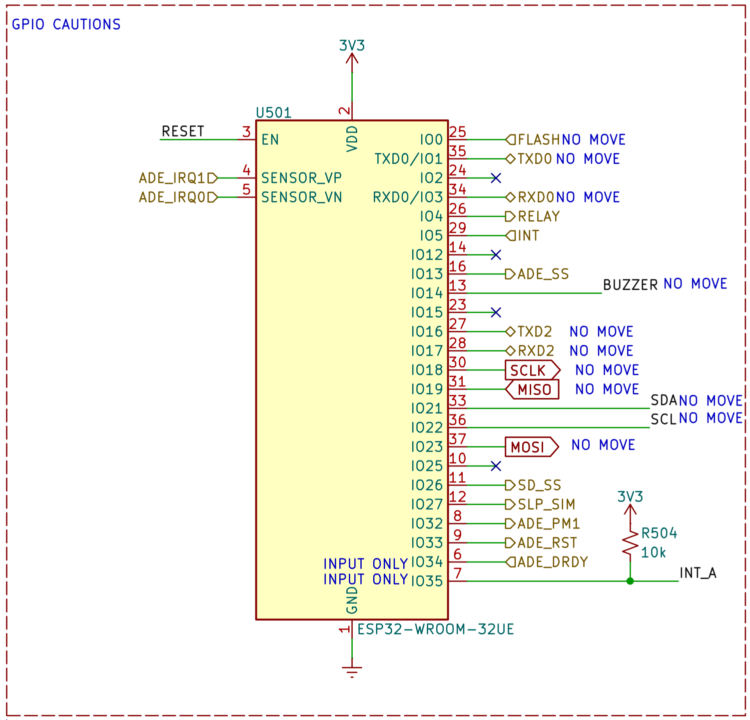
DESIGN NOTE:
Circuit that generates a reset pulse upon power-up, ensuring a clean start once the supply voltage has stabilized.

I2C Pull-Up



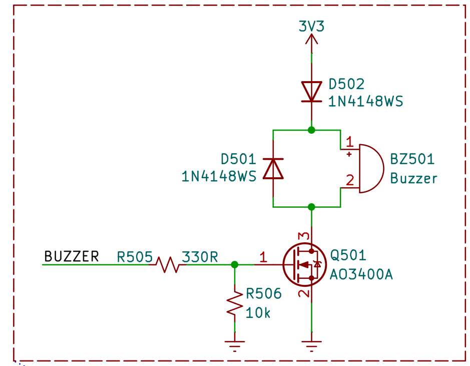
DESIGN NOTE:
Resistors connected between the SDA and SCL lines and the supply voltage are required to ensure proper signal integrity on the I2C bus.

ESP32-WROOM-32E-H4



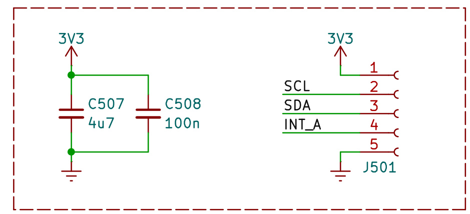
DESIGN NOTE:
The ESP32 offers a dual-core processor that enables efficient processing of multiple tasks, such as energy measurement and simultaneous communication. Its integrated Wi-Fi and Bluetooth connectivity facilitates data transmission to mobile devices and the cloud without the need for additional modules. Furthermore, its low power consumption and flexible I/O with support for multiple protocols (I2C, SPI, UART) make it suitable for portable and energy measurement applications.

Active Buzzer



DESIGN NOTE:
Circuit to control active buzzer that serves as an audible alarm configurable by the user in different situations

2nd PCB



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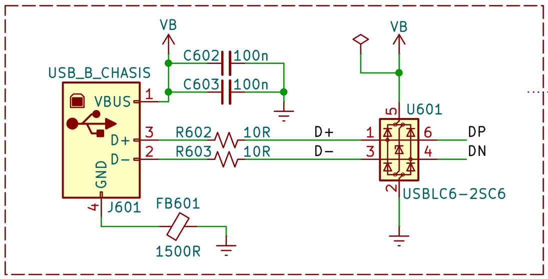
CONFIDENTIAL	PROJECT REVISION: A1	DESIGN ITEM: ESP32 INTERFACE	
APPROVALS / DATE	TITLE: *Power Quality Analyzer ITA-9000*		
ENG: Vazquez Gasca B. D. — 07/21/25	SIZE: USLetter	DWG NO. 25GSK947039-05	REV: 1.0
RV1: Morales Caporal R. — 07/21/25			
RV2:	SCALE: 1:1	FILE NAME: E947039_ESP.kicad_sch	SHEET: 5/13

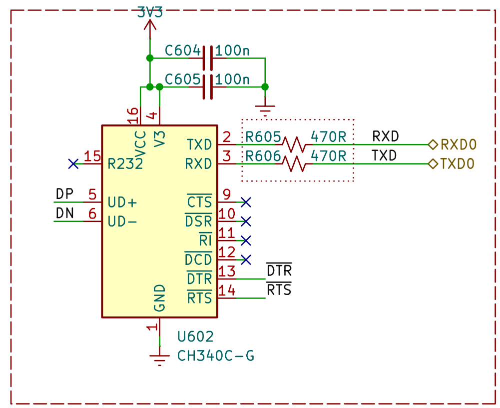
USB-UART INTERFACE

USB-B with ESD Protection

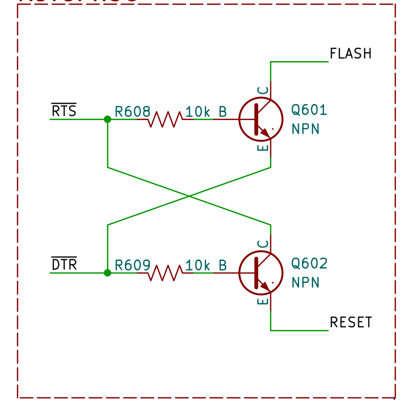


DESIGN NOTE:
The D+ and D- lines of the USB-C connector are protected by series resistors to prevent current spikes, along with a TVS circuit for additional protection against overvoltages and transient spikes in USB communication.

CH340C UART

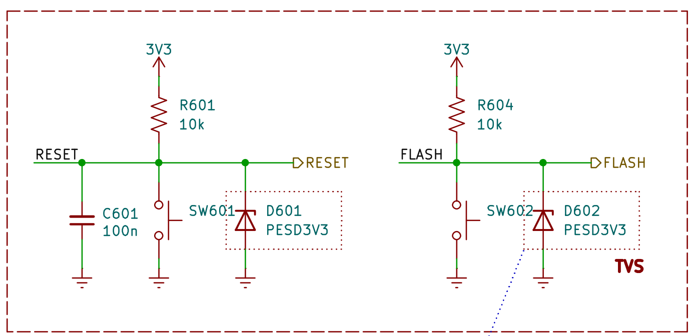


AUTOPROG



DESIGN NOTE:
The USB-UART block has an automatic programming circuit (AUTOPROG), it's not necessary to press any reset or boot button during firmware upload to the ESP32 microcontroller.

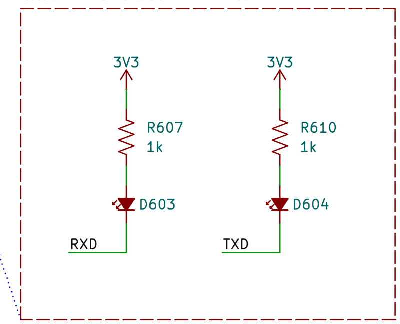
Flash & Reset button



DESIGN NOTE:
Each user-accessible pushbutton has a bi-directional TVS diode for protection against electrostatic discharge up to 30 kV

DESIGN NOTE:
The TX and RX pins of the system have LED indicators for debugging communication errors.

LED Indicator RX & TX



STATUS: PATENT PENDING

CONFIDENTIAL

PROJECT: E947039

TECNOLÓGICO NACIONAL DE MÉXICO

Av. Instituto Tecnológico No. 418
San Andrés Ahuashuatepec Tzomp.
Tlaxcala, México. C.P 90491
Tel: 241-417-2010 Ext.145

PROJECT REVISION: A1
DESIGN ITEM: USB-UART INTERFACE

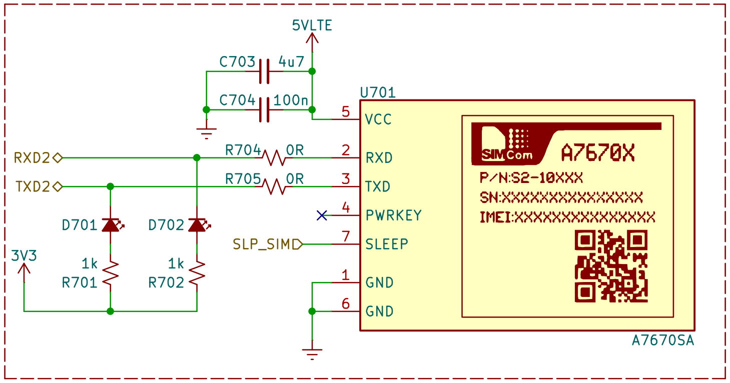
TITLE: Power Quality Analyzer ITA-9000

APPROVALS	DATE
ENG: Vazquez Gasca B. D.	07/21/25
RV1: Morales Caporal R.	07/21/25
RV2:	

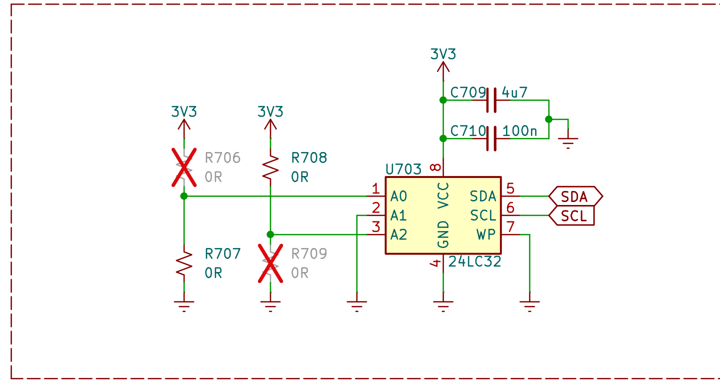
SIZE: USLetter
DWG NO. 25GSK947039-06
REV: 1.0
SCALE: 1:1
FILE NAME: E947039_USB.kicad_sch
SHEET: 6/13

4G LTE, EEPROM, SD CARD, RTC

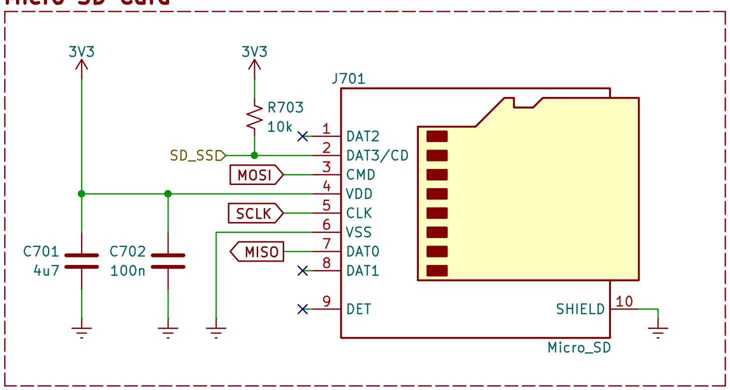
4G LTE Communication



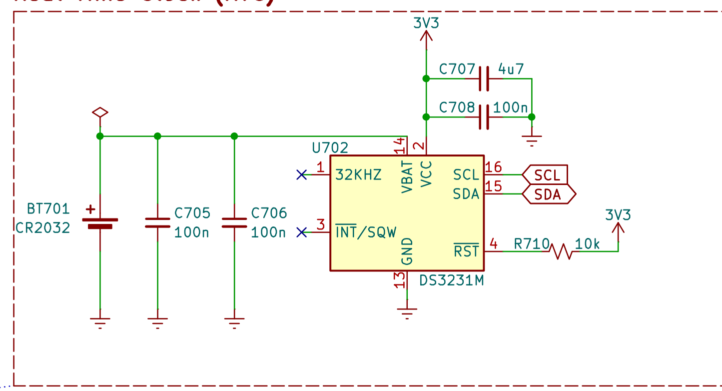
EEPROM 32K bit



Micro SD Card



Real Time Clock (RTC)



DESIGN NOTE:
Removable memory slot used for local storage of measured data when Wi-Fi or 4G LTE cellular connectivity is unavailable, ensuring the integrity and persistence of the collected information.

DESIGN NOTE:
Circuit used to provide an accurate time reference, allowing measured data to be recorded with a date and time stamp, either for local storage or upload to the cloud with reliable time traceability.

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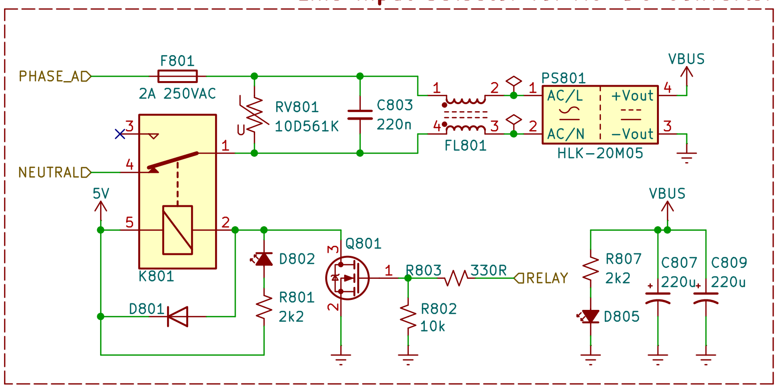
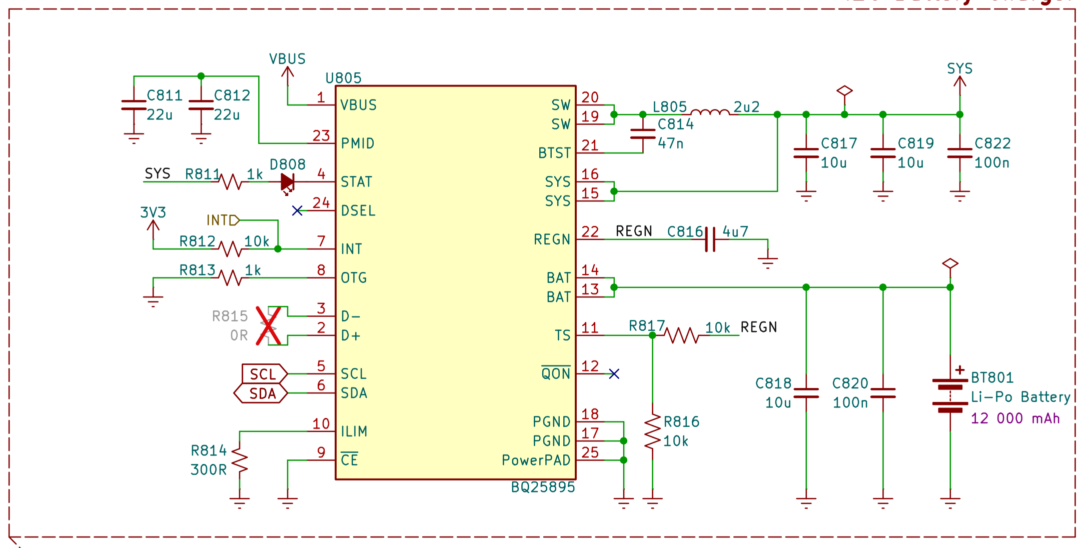
CONFIDENTIAL	PROJECT REVISION: A1	DESIGN ITEM: EEPROM, MICROSD CARD, RTC	
APPROVALS / DATE	TITLE: Power Quality Analyzer ITA-9000		
ENG: Vazquez Gasca B. D. 07/21/25	SIZE: USLetter	DWG NO. 25GSK947039-07	REV: 1.0
RV1: Morales Caporal R. 07/21/25			
RV2:	SCALE: 1:1	FILE NAME: E947039_MEM.kicad_sch	SHEET: 7/13

POWER MANAGEMENT & CHARGING

Line Input Selector for AC–DC Converter

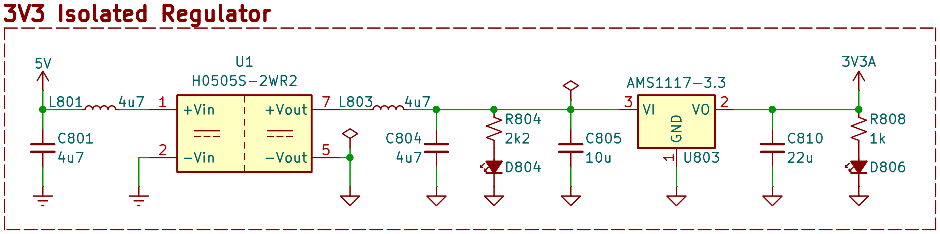


I2C Battery Charger



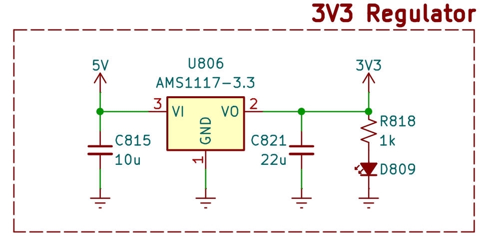
DESIGN NOTE:
Circuit responsible for rapid battery charging and providing a stable DC output, ensuring an efficient and stable supply for powering the system under various operating conditions.

3V3 Isolated Regulator



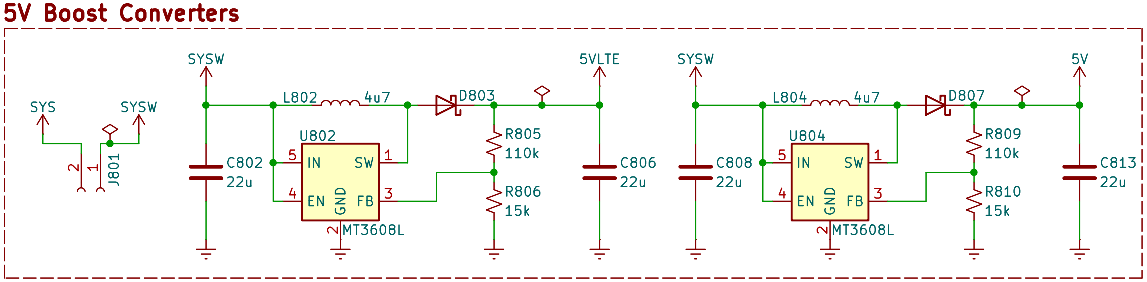
DESIGN NOTE:
Circuit responsible for generating an isolated 3.3V line from a galvanically isolated 5V source. It includes an EMI filter to minimize electromagnetic noise and an LDO regulator that ensures a stable and clean output. This power line is used to energize the integrated circuits located in the high–voltage zone of the system, ensuring safety and electrical separation from the rest of the circuit.

3V3 Regulator



DESIGN NOTE:
LDO regulator that provides a stable 3.3V output to power all digital components in the low voltage area of the logic.

5V Boost Converters



STATUS: PATENT PENDING	PROJECT: E947039	TECNOLÓGICO NACIONAL DE MÉXICO	Av. Instituto Tecnológico No. 418 San Andrés Ahuashuatepec Tzomp. Tlaxcala, México. C.P 90491 Tel: 241–417–2010 Ext.145
CONFIDENTIAL	PROJECT REVISION: A1	DESIGN ITEM: POWER MGMT & CHARGING	
APPROVALS / DATE	TITLE: Power Quality Analyzer ITA–9000		
ENG: Vazquez Gasca B. D. 07/21/25	SIZE: USLetter	DWG NO.: 25GSK947039–08	REV: 1.0
RV1: Morales Caporal R. 07/21/25			
RV2:	SCALE: 1:1	FILE NAME: E947039_PS.kicad_sch	SHEET: 8/13

MECHANICAL DOC

Mounting Holes Main PCB

H901 MountingHole

H904 MountingHole

H902 MountingHole

H903 MountingHole

Fiducial for AOI and PnP

FID901 Fiducial

FID903 Fiducial

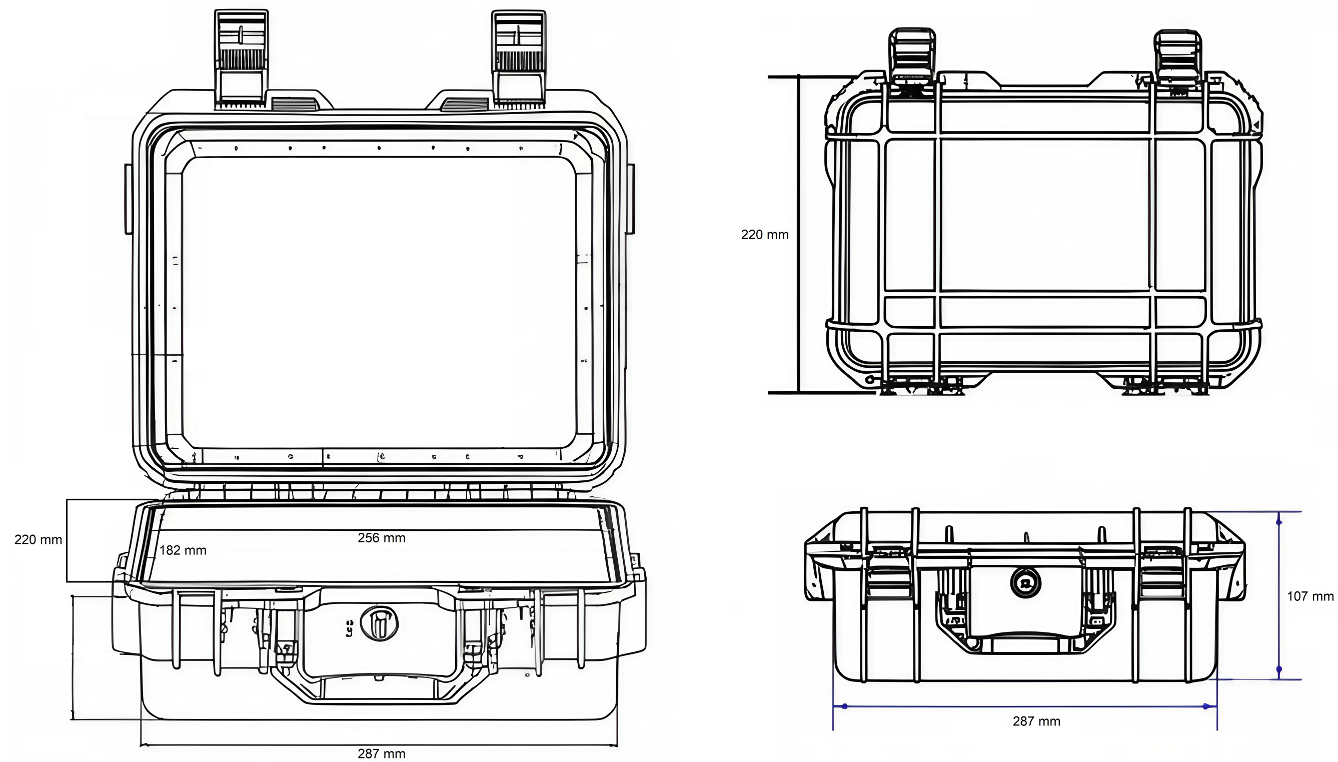
FID905 Fiducial

FID902 Fiducial

FID904 Fiducial

FID906 Fiducial

Enclosure Dimensions



STATUS: PATENT PENDING		PROJECT: E947039	TECNOLÓGICO NACIONAL DE MÉXICO — Av. Instituto Tecnológico No. 418, San Andrés Ahuashuatepec Tzomp. Tlaxcala, México. C.P 90491, Tel: 241-417-2010 Ext.145
CONFIDENTIAL		PROJECT REVISION: A1	DESIGN ITEM: MECHANICAL DOC
APPROVALS	DATE	TITLE: Power Quality Analyzer ITA-9000	
ENG: Vazquez Gasca B. D.	07/21/25	SIZE: USLetter	DWG NO. 25GSK947039-09 — REV: 1.0
RV1: Morales Caporal R.	07/21/25		
RV2:		SCALE: 1:1 — FILE NAME: E947039_MD.kicad_sch	SHEET: 9/13

REVISION HISTORY

12-FEB-2025 DRAFT – Very early stage of schematic, ignore details.

29-APR-2025 PRELIMINARY – Close to final schematic.

03-JUL-2025 CHECKED – There should not be any mistakes. Tell the engineer if you find one.

25-JUL-2025 RELEASED – A board with this schematic has been sent to production.

STATUS: PATENT PENDING	PROJECT: E947039	TECNOLÓGICO NACIONAL DE MÉXICO	Av. Instituto Tecnológico No. 418 San Andrés Ahuashuatepec Tzomp. Tlaxcala, México. C.P 90491 Tel: 241-417-2010 Ext.145
CONFIDENTIAL	PROJECT REVISION: A1	DESIGN ITEM: REVISION HISTORY	
APPROVALS / DATE	TITLE: *Power Quality Analyzer ITA-9000*		
ENG: Vazquez Gasca B. D. — 07/21/25	SIZE: USLetter	DWG NO.: 25GSK947039-A4	REV: 1.0
RV1: Morales Caporal R. — 07/21/25			
RV2:	SCALE: 1:1	FILE NAME: E947039_RH.kicad_sch	SHEET: A4/13